ACCT102 Management Accounting

Week 11

Standard Costs and Variances

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Breaking Down Spending Variances

Isolate changes in Activity

Larry's
Flexible Budge Report
For the Mo 30

	Revenue/Cost Formulas	Planning Budget	Activity Variances	Flexible Budget	Revenue and Spending Variances	Actual Results
Number of lawns (Q)		500		550		550
Revenue	($75Q)	$ 37,500	$ 3,750 F	$ 41,250	$ 1,750 F	$ 43,000
Expenses:						
Wages and salaries	($5,000 + $30Q)	$ 20,000	$ 1,500 U	$ 21,500	$ 2,000 U	$ 23,500
Gasoline and supplies	($9Q)	4,500	450 U	4,950	150 U	5,100
Equipment maintenance	($3Q)	1,500	150 U	1,650	350 F	1,300
Office and shop utilities	($1,000)	1,000	-	1,000	50 F	950
Office and shop rent	($2,000)	2,000	-	2,000	-	2,000
Equipment Depreciation	($2,500)	2,500	-	2,500	-	2,500
Insurance	($1,000)	1,000	-	1,000	200 U	1,200
Total expenses		32,500	2,100 U	34,600	1,950 U	36,550
Net operating income		$ 5,000	$ 1,650 F	$ 6		450

Changes in Price

Changes in Price & Quantity

Spending variances become more useful by breaking them down into **price** and **quantity** variances.

Standard Costs

- Standards are benchmarks or "norms" for measuring performance. In managerial accounting, two types of standards are commonly used.

Quantity standards
specify **how much of an input** should be used to make a product or provide a service.

Price standards
specify **how much should be paid** for each unit of the input.

Quantity standards – Example

Quantity standards specify how much of an input should be used to make a product or provide a service.



Fast-food outlets such as McDonald's have **exacting standards for the quantity of meat going into a sandwich.**



Auto service centers like Firestone and Sears set **labor time standards for the completion of work tasks.**

Price standards – Example

Price standards specify how much should be paid for each unit of the input.

- Hospitals have **standard costs for food, laundry, and other items**.
- Home construction companies have **standard labor costs that they apply to sub-contractors such as framers, roofers, and electricians**.
- Manufacturing companies often have highly developed standard costing systems that establish quantity and price standards for each separate product's material, labor and overhead inputs. These standards are listed on a standard cost card.

Setting Direct Material Standards

Price standards

- Final, delivered cost of materials, net of discounts taken.

Quantity standards

- Amount of material required for each unit of finished product, as well as an allowance for unavoidable waste, spoilage, and other normal inefficiencies.
- Summarized in a Bill of Materials.

Oranges in orange juice: $5 per kg x 0.8 kg

Setting Direct Labor Standards

Price standards

- Includes not only wages earned but also fringe benefits and other labor costs.
- Often a single rate is used that reflects the mix of wages earned.

Quantity standards

- Labor hours required to complete one unit of product.
- Use time and motion studies for each labor operation.

Direct labour in tiling works: $24.5 per hour x 4 hours

Setting Variable Manufacturing Overhead Standards

Price standards

- The rate is the variable portion of the predetermined overhead rate.

Quantity standards

- The quantity is the activity in the allocation base for predetermined overhead.

Variable POHR: $15 per MH x 3 MH

Standard Cost Card – Variable Production Cost

A standard cost card for one unit of product might look like this:

Quantity standards

Price standards

Inputs	A Standard Quantity or Hours	B Standard Price or Rate	A x B Standard Cost per Unit
Direct materials	3.0 lbs.	$ 4.00 per lb.	$ 12.00
Direct labor	2.5 hours	14.00 per hour	35.00
Variable mfg. overhead	2.5 hours	3.00 per hour	7.50
Total standard unit cost			$ 54.50

A General Model for Variance Analysis

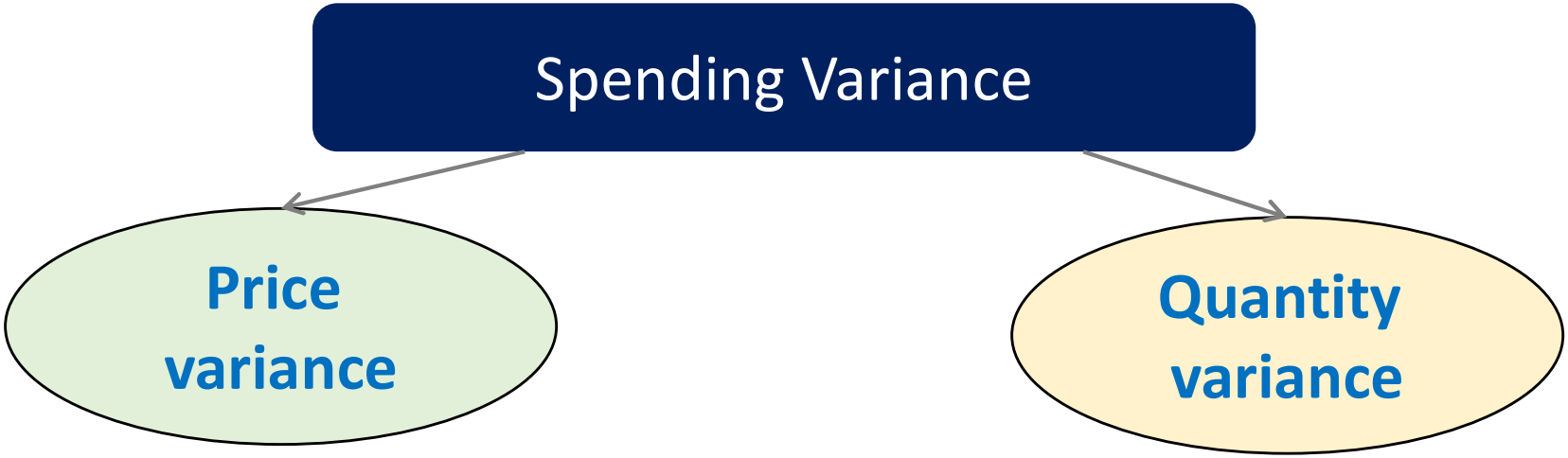
- **Actual quantity (AQ)** is the amount of direct materials, direct labor, and variable manufacturing overhead actually used.

- **Actual price (AP)** is the amount actually paid for the input used.

- **Standard quantity (SQ)** is the standard quantity allowed for the actual output of the period.

- **Standard price (SP)** is the amount that should have been paid for the input used.

A General Model for Variance Analysis



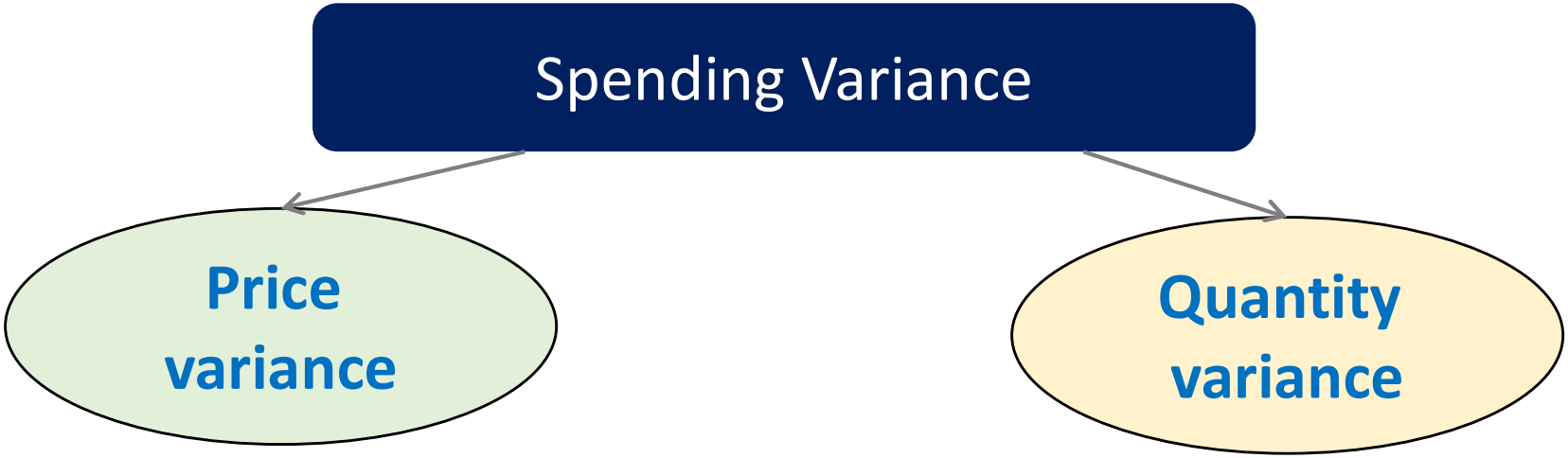
Difference between the actual amount paid for an input and the standard amount that should have been paid.

Difference between how much of an input was actually used and how much should have been used for the actual level of output.

Difference between
actual price
and
standard price

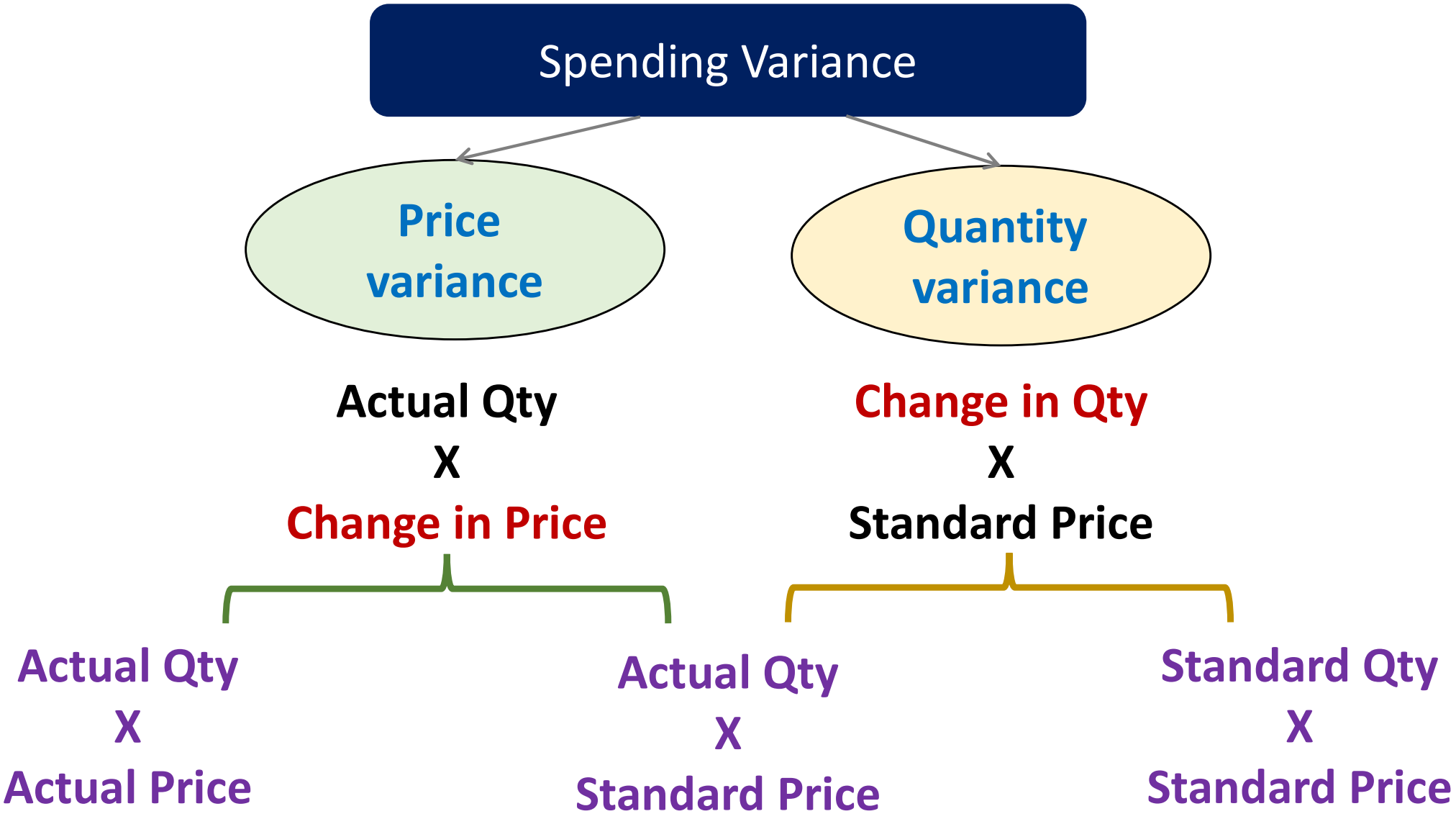
Difference between
actual quantity
and
standard quantity

A General Model for Variance Analysis



Materials price variance	**Direct Materials**	Materials quantity variance
Labor rate variance	**Direct Labor**	Labor efficiency variance
Var. MOH rate variance	**Var. MOH**	Var. MOH efficiency variance

A General Model for Variance Analysis



A General Model for Variance Analysis

AT ACTUAL OUTPUT,

Actual Quantity (AQ)
x **Actual Price (AP)**

Price Variance
(**AQ** x **AP**) – (**AQ** x **SP**)
= **AQ**(**AP** – **SP**)
Change in price x Actual quantity

Actual Quantity (AQ)
x **Standard Price (SP)**

Quantity Variance
(**AQ** x **SP**) – (**SQ** x **SP**)
= **SP**(**AQ** – **SQ**)
Change in quantity x Standard price

Standard Quantity (SQ)
x **Standard Price (SP)**

Learning Objective

Compute the direct materials price and quantity variances and explain their significance.

Material Variances – Example

Glacier Peak Outfitters has the following direct material standard for the fiberfill in its mountain parka.



0.1 kg of fiberfill per parka at $5.00 per kg.



Last month 210 kg of fiberfill were purchased at $4.90 per kg and used to make 2,000 parkas.

Material Variances – Example

Glacier Peak Outfitters has the following direct material standard for the fiberfill in its mountain parka.

0.1 kg of fiberfill per parka at $5.00 per kg.

Last month 210 kg of fiberfill were purchased at $4.90 per kg and used to make 2,000 parkas.

Actual Output?

Actual Price (AP)?

Actual Quantity (AQ)?

Standard Price (SP)?

Standard Quantity (SQ)?

Material Variances – Example

Actual Quantity (AQ) 210kg
x **Actual Price (AP)** x $4.90

$1,029

$21 F

Price Variance
AQ(AP – SP)
210($4.90 – $5)

Actual Quantity (AQ) 210kg
x **Standard Price (SP)** x $5.00

$1,050

$50 U

Quantity Variance
SP(AQ – SQ)
$5(210 – 200)

Standard Quantity (SQ) 200kg
x **Standard Price (SP)** x $5.00

$1,000

Total Variance
= $29 U

0.1 kg per parka × 2,000 parkas = 200 kg

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Quick Check 11B.1

Hanson Inc. has the following direct material standard to manufacture one Zippy:

1.5 pounds per Zippy at $4.00 per pound

Last week, 1,700 pounds of material were purchased and used to make 1,000 Zippies. The material cost a total of $6,630.

Hanson's material **price** variance for the week was:

a. $170 unfavorable.
b. $170 favorable.
c. $800 unfavorable.
d. $800 favorable.

Please attempt to calculate Hanson's material **quantity** variance as well.

Quick Check 11B.1

Hanson Inc. has the following direct material standard to manufacture one Zippy:

1.5 pounds per Zippy at $4.00 per pound

Last week, 1,700 pounds of material were purchased and used to make 1,000 Zippies. The material cost a total of $6,630.

Actual Price (AP)?

Actual Quantity (AQ)?

Standard Price (SP)?

Standard Quantity (SQ)?

Responsibility for Material Variances

Materials Price Variance

Purchasing Manager

Materials Quantity Variance

Production Manager

The **standard** price is used to compute the quantity variance so that the production manager is not held responsible for the purchasing manager's performance.

Responsibility for Material Variances

The materials variances are not always entirely controllable by one person or department.

Your poor scheduling sometimes requires me to rush order material at a higher price, causing unfavorable price variances.



Purchasing Manager

I am not responsible for this unfavorable material quantity variance.

You purchased cheap material, so my people had to use more of it.



Production Manager

Isolation of Material Variances

- Buying and using activities will occur at different times.
- Raw material purchases may be held in inventory for period of time before being used in production.

I need the price variance sooner so that I can better identify purchasing problems.



Purchasing Manager

I'll start computing the **price variance when material is purchased rather than when it's used**.



Management Accountant

- The price variance is computed on the entire quantity **purchased**.
- The quantity variance is computed only on the quantity **used**.

Isolation of Material Variances

- The price variance is computed on the entire quantity **purchased**.
- The quantity variance is computed only on the quantity **used**.

EXAMPLE: Hanson had higher purchases of material then material used.

Hanson Inc. has the following material standard to manufacture one Zippy:

1.5 pounds per Zippy at $4.00 per pound

Last week, **2,800 pounds of material were purchased at $3.90 per pound**, and 1,700 pounds were used to make 1,000 Zippies.

Isolation of Material Variances

Actual Quantity (AQ)

x **Actual Price (AP)**



The price variance is computed on the entire quantity **purchased**.

Actual Quantity (AQ)

x **Standard Price (SP)**



The quantity variance is computed only on the quantity **used**.

Standard Quantity (SQ)

x **Standard Price (SP)**

Material Price Variance – Example

Actual Quantity (AQ)

PURCHASED

x **Actual Price (AP)**

2,800lbs

x **$3.90**

$10,920

$280 F

Price Variance

AQ(AP – SP)

2,800($3.90 – $4)

Actual Quantity (AQ)

PURCHASED

x **Standard Price (SP)**

2,800lbs

x **$4.00**

$11,200

Price variance increases because quantity purchased increases.

Material Quantity Variance – Example

Quantity variance is unchanged because actual and standard quantities are unchanged.

Actual Quantity (AQ)

USED

x **Standard Price (SP)**

1,700lbs

x **$4.00**

$6,800

Quantity Variance

SP(AQ – SQ)

$800 U

$4(1,700 – 1,500)

Standard Quantity (SQ)

x **Standard Price (SP)**

1,500lbs

x **$4.00**

$6,000

1.5 lbs × 1,000 Zippy = 1,500 lbs

Learning Objective

Compute the direct labor rate and efficiency variances and explain their significance.

Labor Variances – Example

Glacier Peak Outfitters has the following direct labor standard for its mountain parka.

1.2 standard hours per parka at $10.00 per hour



Last month, employees actually worked 2,500 hours at a total labor cost of $26,250 to make 2,000 parkas.



Labor Variances – Example

Glacier Peak Outfitters has the following direct labor standard for its mountain parka.

1.2 standard hours per parka at $10.00 per hour

Last month, employees actually worked 2,500 hours at a total labor cost of $26,250 to make 2,000 parkas.

Actual Output?

Actual Price (AP)?

Actual Quantity (AQ)?

Standard Price (SP)?

Standard Quantity (SQ)?

Labor Variances – Example

Actual Quantity (AQ)

x **Actual Price (AP)**

Actual Quantity (AQ)

x **Standard Price (SP)**

Standard Quantity (SQ)

x **Standard Price (SP)**

Labor Variances – Example

$1,029 ÷ 210 kgs = $4.90 per kg

Actual Quantity (AQ) **2,500hrs**
x **Actual Price (AP)** x **$10.50**

$26,250

$1,250 U

Rate Variance
AQ(AP – SP)
2,500($10.50 – $10)

Actual Quantity (AQ) **2,500hrs**
x **Standard Price (SP)** x **$10**

$25,000

$1,000 U

Efficiency Variance
SP(AQ – SQ)
$10(2,500 – 2,400)

Standard Quantity (SQ) **2,400hrs**
x **Standard Price (SP)** x **$10**

$24,000

1.2 hrs x 2,000 parkas = 2,400 hrs

Total Variance
= $2,250 U

Responsibility for Labor Variances

Production managers are usually held accountable for labor variances because they can influence the:

- Mix of skill levels assigned to work tasks.
- Level of employee motivation.
- Quality of production supervision.
- Quality of training provided to employees.

Responsibility for Labor Variances

Labor variances are not entirely controllable by one person or department.

I am not responsible for the unfavorable labor efficiency variance!

You purchased cheap material, so it took more time to process it.



Production Manager

I think it took more time to process the materials because the Maintenance Department has poorly maintained your equipment.

Learning Objective

Calculations not examinable

Compute the variable manufacturing overhead rate and efficiency variances.

Variable Manufacturing Overhead Variances – Example

Glacier Peak Outfitters has the following direct variable manufacturing overhead standard for its mountain parka.

Allocation base : DLH

1.2 standard hours per parka at $4.00 per hour

Last month, employees actually worked 2,500 hours to make 2,000 parkas. Actual variable manufacturing overhead for the month was $10,500.





Variable Manufacturing Overhead Variances – Example

$10,500 ÷ 2,500 hours = $4.20 per hour

Actual Quantity (AQ) **2,500hrs**
x **Actual Price (AP)** x **$4.20** **$10,500**

$500 U

Rate Variance
AQ(AP – SP)
2,500($4.20 – $4)

Actual Quantity (AQ) **2,500hrs**
x **Standard Price (SP)** x **$4** **$10,000**

$400 U

Efficiency Variance
SP(AQ – SQ)
$4(2,500 – 2,400)

Standard Quantity (SQ) **2,400hrs**
x **Standard Price (SP)** x **$4** **$9,600**

1.2 hours per parka x 2,000 parkas = 2,400 hours

Total Variance
= $900 U

Variance Analysis and Management by Exception

How do I know which variances to investigate?

Larger variances, in dollar amount or as a percentage of the standard, are investigated first.

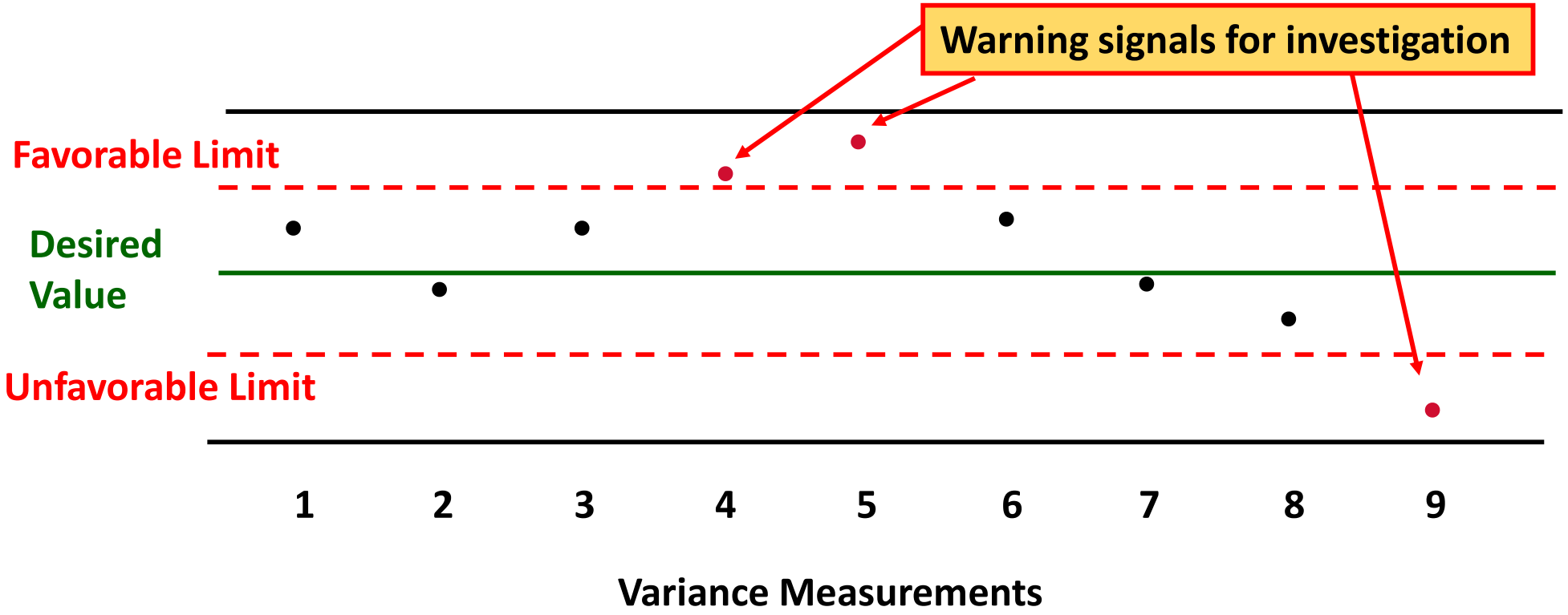
E.g. Pasta manufacturer

Price variance	**$**	**% from SP**
DM – Flour	1,500,000 U	1% U
DM - Salt	10,000 U	5% U
DM – Oil	150,000 U	40% U

A Statistical Control Chart

Plotting variance analysis data on a statistical control chart is helpful in variance investigation decisions. Variances are investigated if:

- They are unusual relative to the normal level of random fluctuation.
- An unusual pattern emerges in the data.

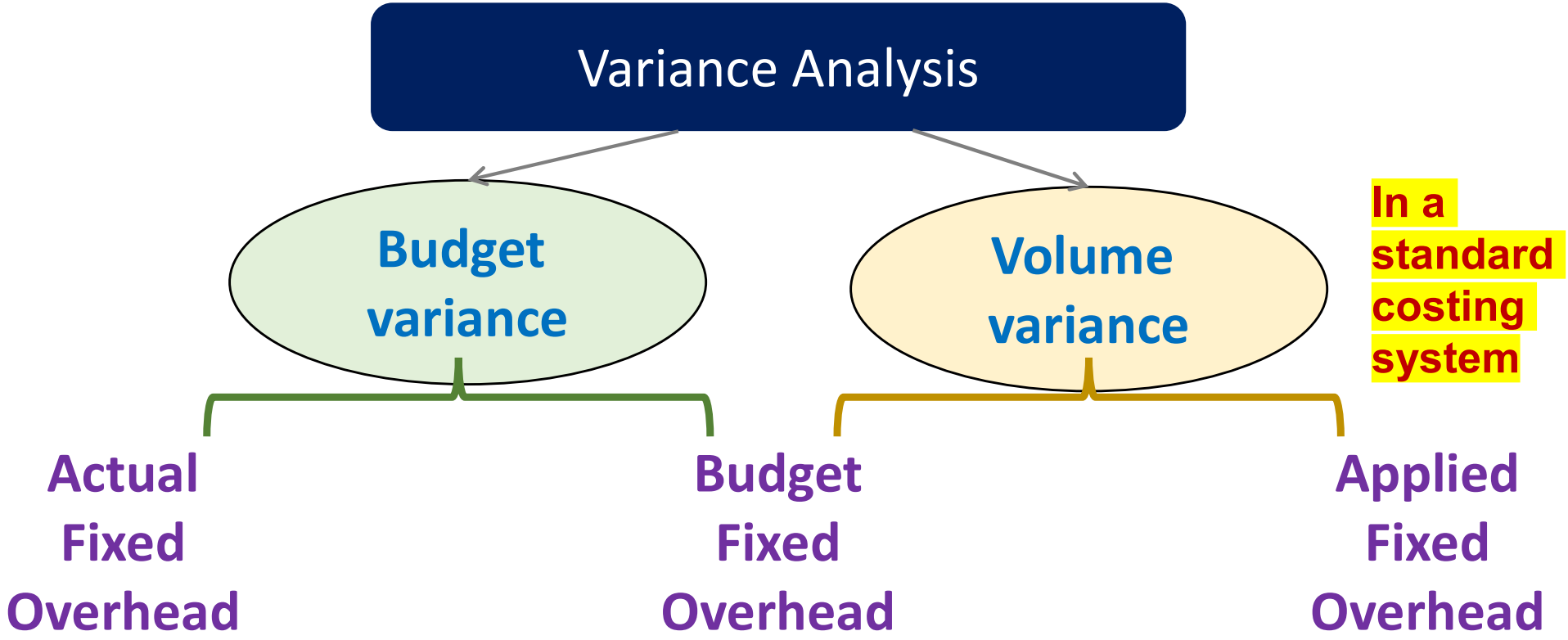


Learning Objective

Predetermined Overhead Rates and Overhead Analysis in a Standard Costing System

Compute and interpret the fixed overhead budget and volume variances.

Fixed Overhead Variance



Actual FOH – Budget FOH = Budget Variance

Budget FOH – Applied FOH = Volume Variance

Fixed Overhead Variance

Isolate changes in Activity

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Flexible Budge Report
For the Mo 30

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Equipment maintenance	($3Q)	1,500	150 U	1,650	350 F	1,300
Office and shop utilities	($1,000)	1,000	-	1,000	50 F	950
Office and shop rent	($2,000)	2,000	-	2,000	-	2,000
Equipment Depreciation	($2,500)	2,500	-	2,500	-	2,500
Insurance	($1,000)	1,000	-	1,000	200 U	1,200
Total expenses		32,500	2,100 U	34,600	**Budget variance**	36,550
Net operating income		$ 5,000	$ 1,650 F	$ 6,650		6,450

Applied Fixed Overhead in Variance Analysis

Note the difference between normal costing application and standard costing application.

STANDARD costing

Applied Fixed Overhead = Predetermined overhead rate × **Standard hours/quantity** allowed for the **actual output**

Previously, what we learnt under **NORMAL costing**

Applied Fixed Overhead = Predetermined overhead rate × **Actual activity (Actual hours/quantity used** for the **actual output**)

Fixed Overhead Volume Variance

- Volume variance is favorable if the actual level of activity is greater than planned
- Does not measure overspending or underspending, **viewed more as a measure of utilization of facilities.**

Budget Fixed Overhead > **Applied Fixed Overhead**

DH > SH

Unfavourable Volume Variance

Budget Fixed Overhead < **Applied Fixed Overhead**

DH < SH

Favourable Volume Variance

Fixed Overhead Variance – Example (1)

Budgeted production	**10,000**	**units**
Budgeted Fixed manufacturing overheads	**$150,000**	
Actual production	**12,000**	**units**
Actual Fixed manufacturing overheads	**$160,000**	

Actual Fixed Overhead	Budget Fixed Overhead	Applied Fixed Overhead
$160,000	**$150,000**	**$180,000**
		$150k / 10k * 12k

Budget var. = $10k U

Volume var. = $30k F

Fixed Overhead Variance – Example (2)

Budgeted fixed manufacturing overhead	**$270,000**
Budgeted production	**30,000**
Standard machine-hour per unit	**3**
Budgeted machine-hour	**90,000**
Actual fixed manufacturing overhead	**$280,000**
Actual production	**28,000**
Actual machine-hour	**88,000**
Standard machine-hour allowed for the actual production	**84,000**

3 SH per unit x 28,000 units = 84,000 SH

Predetermined overhead rate = $270,000 / 90,000 MH

= $3 per MH

Fixed Overhead Variance – Example (2)

Actual FOH

Budget FOH **Denominator Hours (DH)**

x **FPOHR**

Budget Variance

Applied FOH **Standard Hours (SH)**

x **FPOHR**

Volume Variance

(DH x FPOHR) –

(SH x FPOHR)

= FPOHR(DH – SH)

FPOHR = Fixed portion of the predetermined overhead rate

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Fixed Overhead Variance – Example (2)

Actual FOH **$280,000**

$10,000 U **Budget Variance**

Budget FOH

Denominator Hours (DH) **90,000hrs**

x **FPOHR (FR)** x **$3**

$270,000

Volume Variance

FPOHR(DH – SH)

$3(90,000 – 84,000)

$18,000 U

Applied FOH

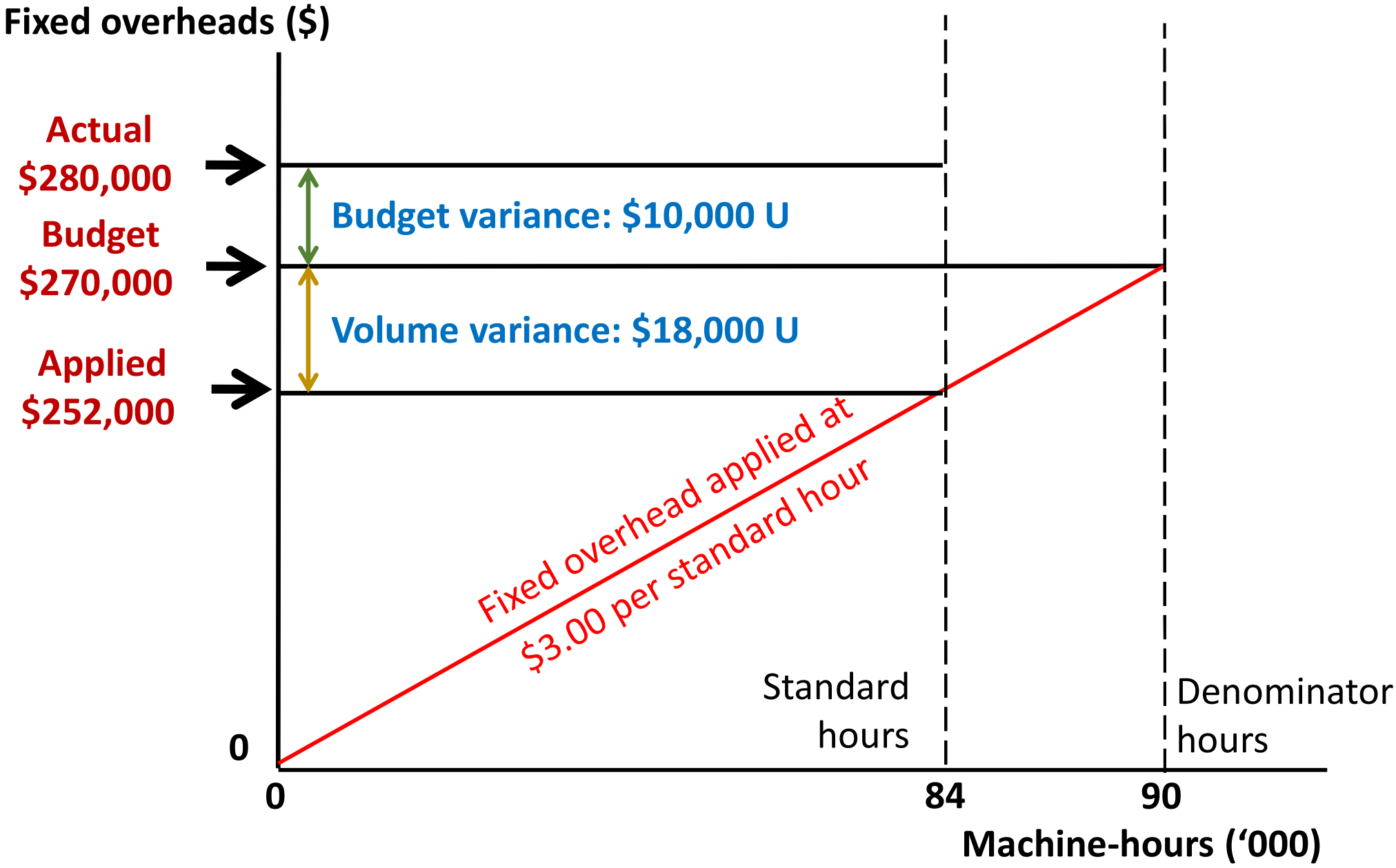
Standard Hours (SQ) **84,000hrs**

x **FPOHR (SP)** x **$3**

$252,000

Total Variance = $28,000 U

Graphic Analysis of Fixed Overhead Variances



End of topic